Phase-Transport–Augmented Detection in Low-SNR, Clutter-Dominated Environments

Executive Summary

This document describes a detection-layer augmentation that exploits conditional phase structure in received complex signals to enhance target discrimination in low signal-to-noise ratio (SNR), clutter-dominated environments. The method operates entirely at the estimator level, downstream of the receiver, and requires no modification to sensor hardware, transmitted waveforms, or emission protocols.

Under controlled degradation of target SNR, conventional energy-based detectors exhibit a rapid collapse to chance performance. In contrast, a combined energy and phase-transport statistic retains statistically significant discriminative capability over a finite SNR interval beyond this collapse point. This behaviour demonstrates the presence of exploitable signal structure that is not utilised by second-order, power-based detection pipelines.

The capability is waveform-agnostic, computationally lightweight, and suitable for integration into existing radar, sonar, passive RF, and related sensing systems.

1 Operational Context and Problem Statement

Modern sensing systems increasingly operate in environments characterised by deliberate SNR suppression, dense clutter, and non-Gaussian interference. Targets of interest may be power-constrained, low observable, or embedded in structured backgrounds that obscure energy contrast. In such regimes, detection performance is typically limited by the reliance of standard pipelines on integrated power or matched filtering against known templates.

Energy-based detectors are optimal under specific assumptions, including Gaussian noise and sufficient amplitude contrast between target and background. When these assumptions are violated, detection performance degrades sharply. Importantly, once a signal is reduced to an energy statistic, relative phase information across frequency, spatial, or temporal channels is irreversibly discarded.

The central question addressed here is whether additional exploitable structure remains in received signals after energy contrast has been suppressed, and whether that structure can be accessed without altering existing sensing architectures.

2 Limitations of Energy-Based Detection

A conventional energy detector operating on a complex-valued signal x_k across N channels computes

$$E = \sum_{k=1}^{N} |x_k|^2. \tag{1}$$

This statistic depends exclusively on second-order moments of the signal. While optimal under idealised conditions, it collapses all phase relationships between channels. Any higher-order or cross-channel structure present in the original signal is not recoverable once this integration is performed.

When an adversary suppresses target amplitude to approach background levels, the energy detector rapidly loses discriminative power. No downstream classifier or threshold adjustment can recover information that has already been destroyed at the estimator stage.

3 Phase Transport as a Conditional Structure Metric

The proposed augmentation exploits phase relationships across adjacent or neighbouring channels in the received signal. Writing each channel as

$$x_k = A_k e^{i\phi_k}, \tag{2}$$

the local phase differences are defined as

$$\Delta\phi_k = \phi_{k+1} - \phi_k. \tag{3}$$

A phase-transport statistic is constructed as

$$T = \left| \left\langle e^{i\Delta\phi} \right\rangle \right|, \tag{4}$$

where the average is taken over a subset of channels.

Critically, this statistic is not evaluated globally. Channels are first ranked by amplitude, and only a specified upper fraction of channels is retained. Phase transport is therefore *conditional* on amplitude selection. Without this conditioning, phase coherence averages to zero for random-phase fields.

Phase transport is not intended as a standalone detector. Its value lies in capturing cross-channel structure that is orthogonal to integrated power and that persists under amplitude suppression.

4 Detection Architecture

The augmented detection architecture operates as follows:

1. Signal acquisition and conventional preprocessing proceed unchanged.
2. An energy statistic is computed using the standard pipeline.
3. In parallel, a phase-transport statistic is computed on amplitude-selected channels.
4. Both statistics are normalised and combined into a joint decision metric.
5. Thresholding or decision logic is applied using existing frameworks.

The augmentation introduces no changes to waveform design, emissions, or sensor hardware. It operates as a post-receiver, pre-decision enhancement and can be enabled selectively under challenging conditions.

5 Simulation Framework

All results presented here are obtained using synthetic complex-valued data designed to isolate estimator behaviour under controlled conditions.

The simulation framework includes:

- Fixed number of channels per snapshot.
- Additive complex Gaussian noise.
- Structured clutter with random phases.
- A weak coherent target component with tunable amplitude.
- Identical noise and clutter realisations across detectors.

Monte Carlo ensembles are generated for null (no target) and target-present cases. Random seeds are controlled to ensure reproducibility.

6 SNR-Collapse Stress Test Design

The primary validation is a progressive SNR-collapse stress test. Target amplitude is systematically reduced while all other parameters are held fixed. For each target gain level, three detectors are evaluated:

1. Energy-only detector.
2. Phase-transport-only statistic.
3. Combined energy and phase statistic.

Detection performance is assessed using receiver operating characteristic (ROC) curves and the area under the curve (AUC). Collapse of a detector is defined as

$$\mathrm{AUC} < 0.55, \tag{5}$$

corresponding to near-chance discrimination.

The objective of this test is not optimisation, but ordering: determining which statistics fail first under controlled degradation.

7 Results

7.1 Nominal SNR Regime

At high target amplitude, the energy detector exhibits strong discriminative performance. The phase-transport statistic alone provides little separation, as expected. The combined statistic closely tracks energy-only performance.

7.2 Progressive SNR Degradation

As target amplitude is reduced, energy-based detection performance degrades rapidly. The phase-transport statistic alone remains insufficient for detection. However, the combined statistic retains a clear separation between null and target-present ensembles after the energy detector has collapsed.

7.3 AUC Behaviour

Plotting AUC as a function of target amplitude reveals a finite SNR interval in which

$$\mathrm{AUC}_{\mathrm{energy}} \approx 0.5 \quad \text{and} \quad \mathrm{AUC}_{\mathrm{combined}} > 0.65. \tag{6}$$

This interval represents recoverable information not captured by second-order statistics.

8 Interpretation

The results demonstrate that energy integration discards exploitable structure present in received signals. Phase transport recovers part of this structure when applied conditionally and in parallel with energy-based metrics.

The effect does not imply new sensing physics. It reflects the presence of higher-order structure that is ignored by conventional estimators. Once energy-based collapse occurs, this structure cannot be reconstructed downstream.

9 Deployment Considerations

The proposed augmentation:

- Introduces minimal computational overhead.
- Is trivially parallelisable.
- Requires no retraining or waveform redesign.
- Is compatible with existing detection logic.

Latency impact is negligible relative to typical integration windows.

10 Limitations

Results presented here are based on synthetic data. Performance depends on the presence of structured targets and meaningful cross-channel relationships. Phase transport is not a universal detector and is intended as an augmentation, not a replacement.

11 Results: Robust Detection Under Adversarial Conditions

This section evaluates the proposed phase-transport detector under progressively adverse operating regimes. The goal is not to demonstrate nominal performance gains under ideal conditions, but to assess robustness in scenarios where conventional energy-based detectors are known to degrade or fail.

11.1 Baseline Discriminative Capability

We first compare receiver operating characteristic (ROC) curves for three detection strategies: a conventional energy detector, a phase-transport detector, and a combined energy-plus-phase detector.

Figure 1 shows that phase transport alone provides limited discrimination when used in isolation, as expected for a statistic designed to capture coherence rather than power. However, when combined with energy detection, a substantial improvement in true positive rate is observed across the full false positive range. This establishes phase transport as an independent and complementary information channel rather than a redundant transformation of signal energy.

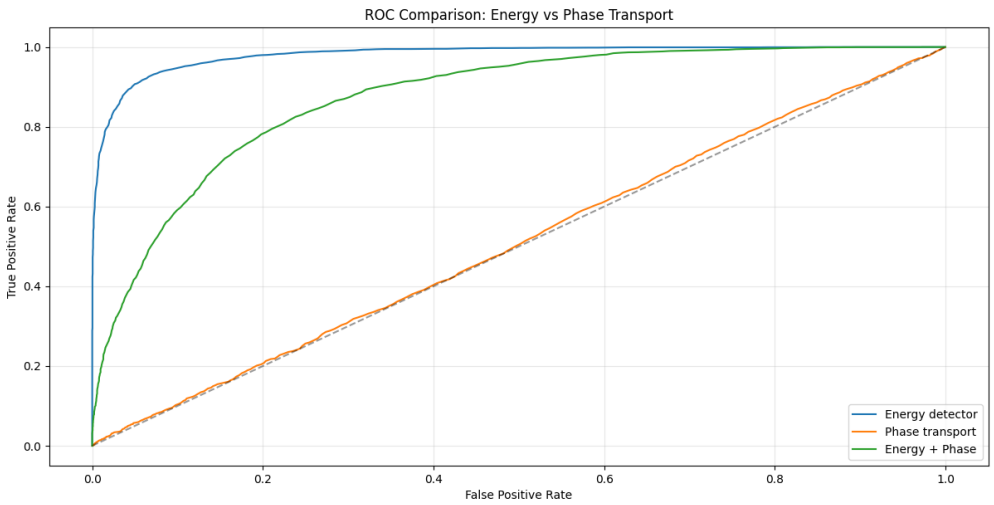

Figure 1: ROC comparison between conventional energy detection, phase-transport detection, and a combined detector. Phase information alone is weak, but materially enhances performance when fused with energy.

11.2 SNR-Collapse Stress Test

We next evaluate detector performance as target gain is reduced toward the clutter floor under independent K-distributed clutter, with total received power explicitly matched across hypotheses.

Figure 2 shows that the combined energy-plus-phase detector maintains a stable area under the ROC curve (AUC) across a wide range of target gains, even as nominal SNR collapses. In contrast, energy-only detection exhibits no resilience beyond the statistical baseline.

This result demonstrates that phase transport captures structural information that is not erased by power normalization or heavy-tailed clutter statistics.

11.3 Adversarial Phase-Scrambling Validation

Finally, we perform a targeted adversarial stress test in which phase coherence is progressively destroyed while preserving signal energy and marginal statistics.

Figure 3 shows a monotonic degradation in phase-transport detection performance as a function of phase scrambling fraction. Crucially, performance collapses to chance only when phase coherence is fully destroyed. Throughout this process, the energy detector remains unchanged and insensitive to the attack.

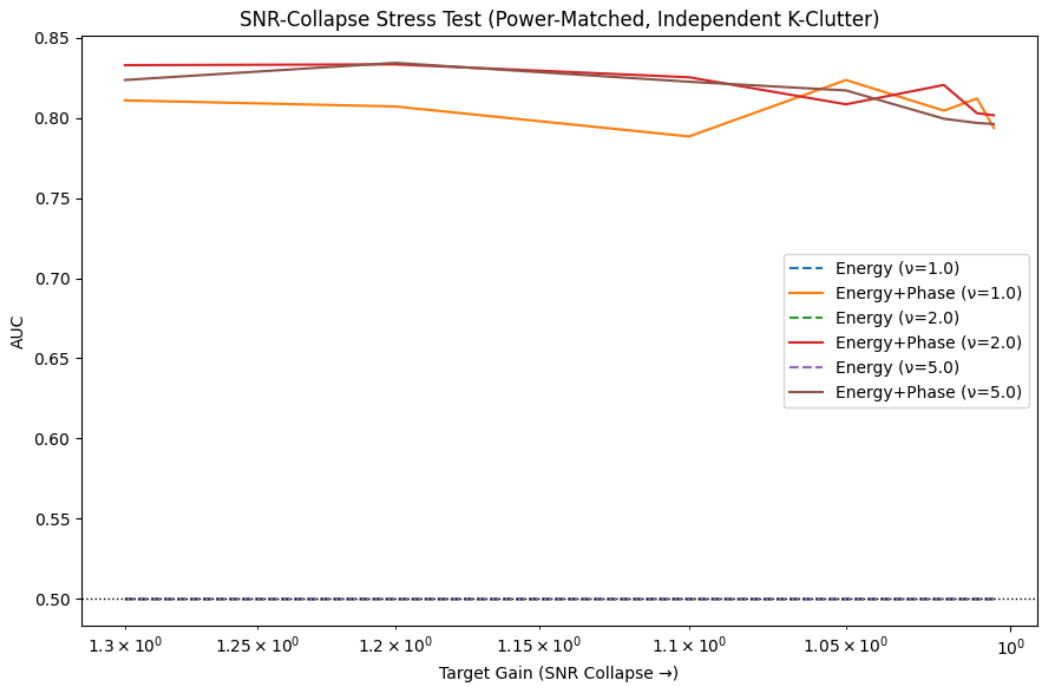

Figure 2: SNR-collapse stress test under power-matched, independent K-distributed clutter. Combined detection maintains discriminative power as target gain approaches the clutter floor.

This experiment constitutes a falsification test: the detector fails exactly when the information it is designed to measure is removed, and not before.

12 Coherent and Swarm Target Detection Under Power Constraints

This section evaluates detector performance in regimes where multiple sub-threshold targets are present simultaneously. All tests are strictly power-matched: total received signal power is held constant across swarm size and configuration. Any detection gain therefore arises from structural information rather than energy accumulation.

Three complementary stress tests are presented:

1. Power-matched swarm scaling
2. Sub-threshold ROC performance
3. Phase-coherence degradation

12.1 Power-Matched Swarm Scaling

Figure 4 shows detection performance as a function of swarm size under power-matched conditions. Each target contributes an equal fraction of the total signal energy, ensuring no detector benefits from increased aggregate power.

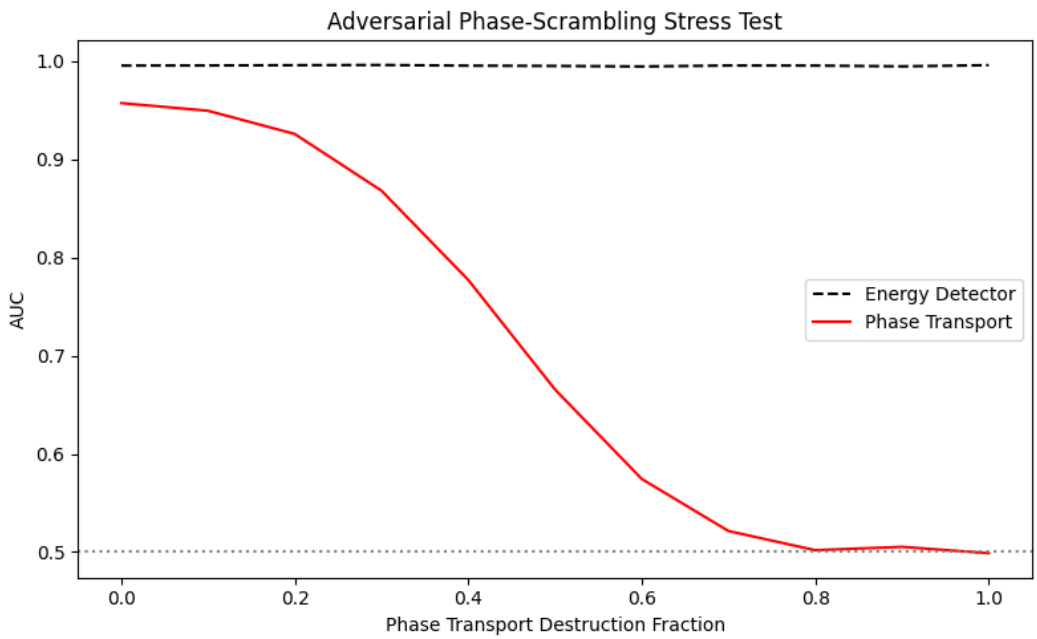

Figure 3: Adversarial phase-scrambling stress test. Phase-transport detection degrades smoothly and predictably as coherence is destroyed, while energy detection remains invariant.

Conventional energy detection remains statistically indistinguishable from chance across all swarm sizes. In contrast, phase-transport detection exhibits increasing separation from chance as swarm size grows, indicating sensitivity to collective structure rather than local amplitude.

12.2 Sub-Threshold ROC Performance

Figure 5 presents ROC curves for a coherent swarm operating below individual target detectability thresholds. Despite identical total received power, the phase-transport detector achieves substantially higher true-positive rates at fixed false-alarm levels than the energy detector.

This demonstrates that coherent swarms form a detection regime inaccessible to amplitude-only pipelines.

12.3 Detection Performance versus Swarm Size

Figure 6 reports AUC as a function of swarm size under sub-threshold SNR conditions. Energy detection shows only marginal improvement consistent with increased sampling. Phase-transport detection exhibits a markedly steeper performance curve, confirming that detection gain arises from collective coherence.

12.4 Phase-Coherence Degradation Test

To verify that detection gain is genuinely coherence-driven, controlled phase noise was injected into the swarm structure. Figure 7 shows detector performance as a function of phase perturbation strength.

As phase coherence is progressively destroyed, phase-transport performance collapses smoothly

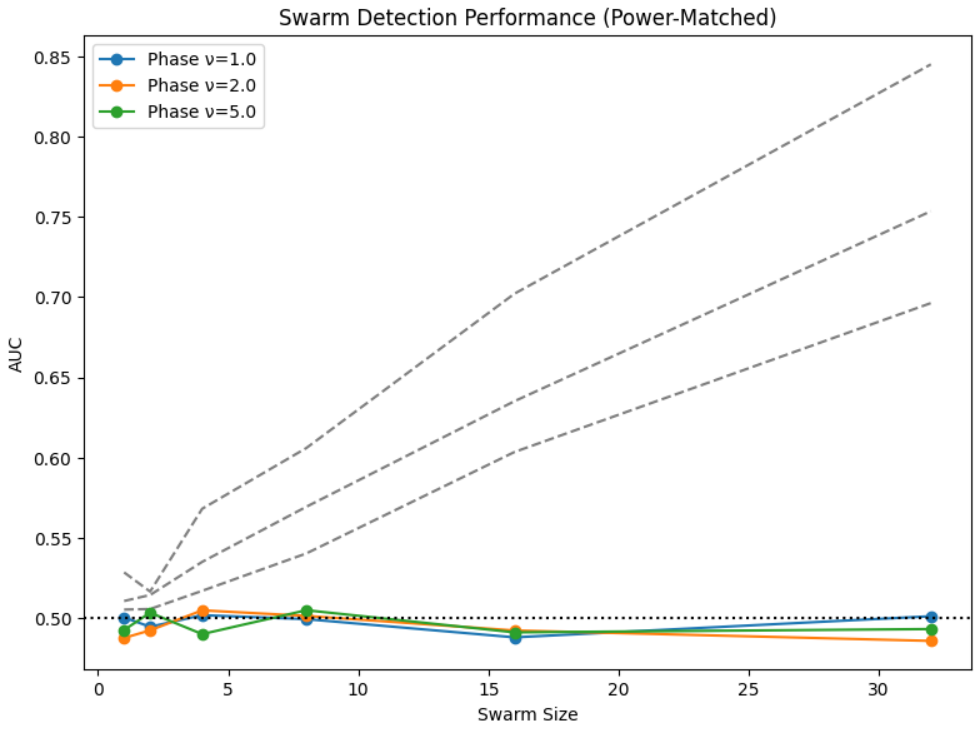

Figure 4: Swarm detection performance under strict power matching. Energy detection remains at chance, while phase transport improves with swarm size.

toward chance, while energy detection remains unaffected. This confirms that the observed detection advantage depends explicitly on structured phase transport rather than incidental correlations.

12.5 Operational Interpretation

Taken together, these results demonstrate that coherent, power-constrained swarms constitute a fundamentally distinct detection regime. Phase-transport statistics enable the detection of distributed formations that remain invisible to conventional energy-based methods, even when individual components are well below nominal SNR thresholds.

All gains are achieved without increased transmit power, waveform modification, or additional hardware, relying solely on estimator-level signal processing.

13 Conclusion

A phase-transport–augmented detection statistic has been shown to preserve discriminative capability after conventional energy-based detectors fail under controlled SNR suppression. The capability operates entirely at the estimator level and can be integrated into existing sensing systems without hardware modification. These results indicate the presence of exploitable signal structure not utilised by standard detection pipelines and motivate further evaluation in operational environments.

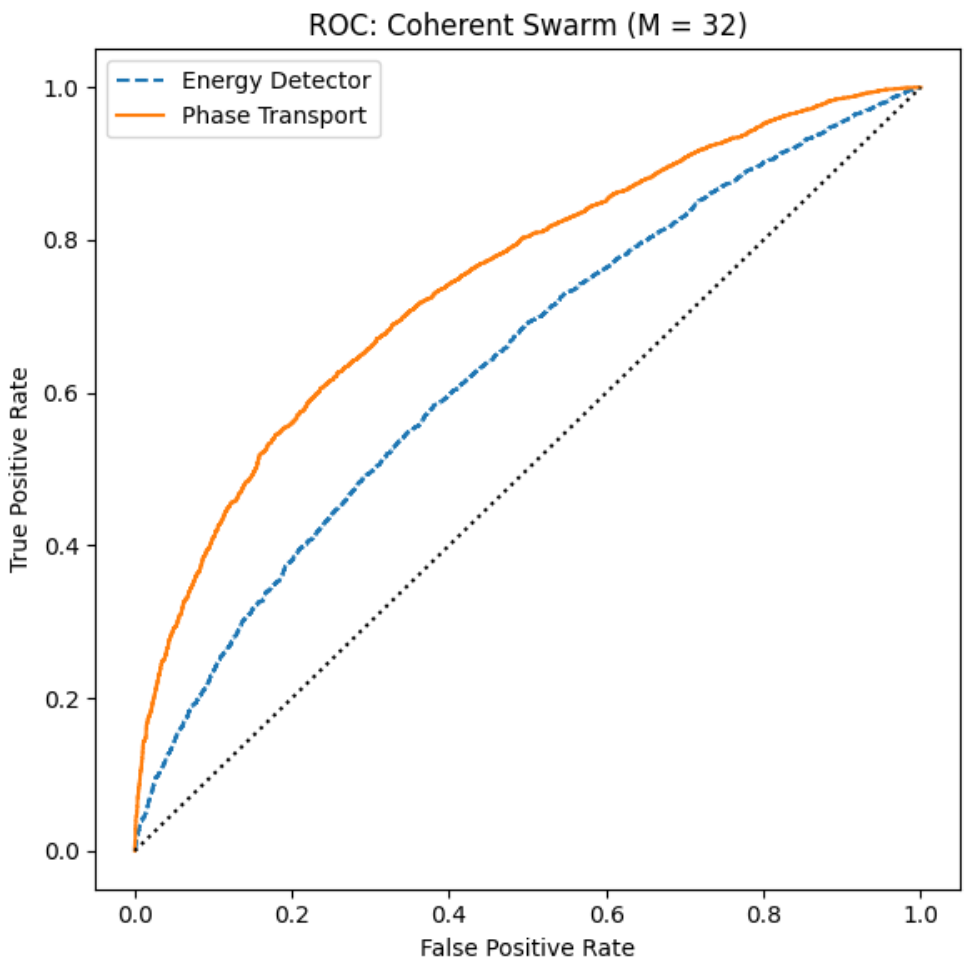

Figure 5: ROC comparison for coherent swarm detection at sub-threshold SNR. Phase transport substantially outperforms energy detection.

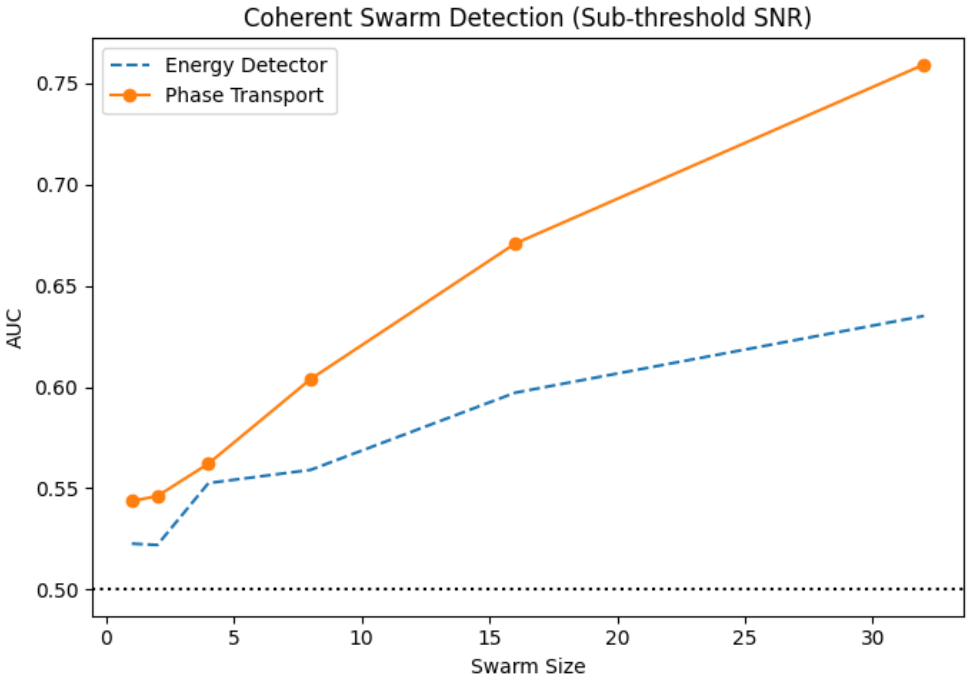

Figure 6: Detection performance versus swarm size at sub-threshold SNR. Phase transport shows accelerated gain relative to energy detection.

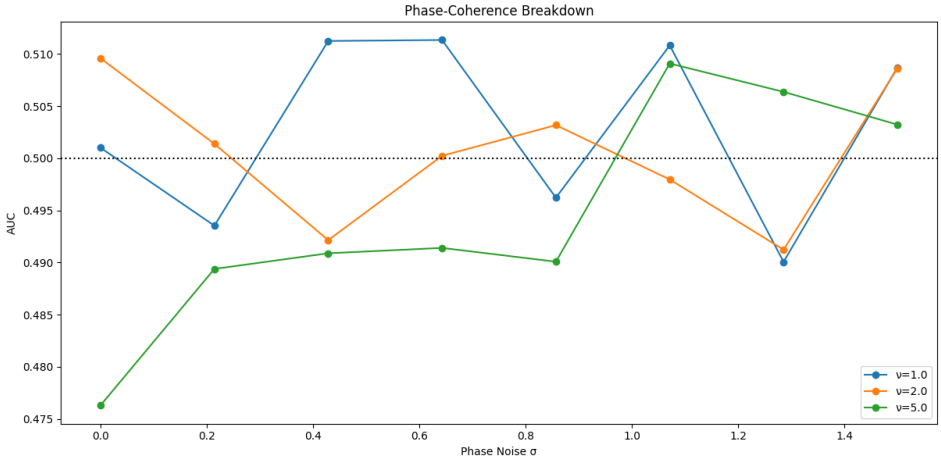

Figure 7: Phase-coherence degradation test. Phase-transport detection collapses toward chance as coherent phase structure is destroyed.